Annotx Technical Assessment

Automated Entity Recognition & Interactive Feature Visualizer for Embedded Systems Specifications

CANDIDATE NAME	BACKEND ENGINE	STATUS
Deepak V	**Python (Regex / JSON)**	**Complete & Verified**
ASSESSMENT DATE	FRONTEND	
October 16, 2026	**HTML5 / JavaScript / Tailwind**	

1 BACKEND EXTRACTION LOGIC (PYTHON)

The backend engine (**Extractor.py**) parses unstructured strings against target taxonomies across four core domain categories: **Hardware, Software, Protocols** and **Languages**. Regular expressions enforced with strict word boundaries (\b) prevent partial matches (e.g., preventing the language 'C' from matching the character 'c' inside general words).

```
import re, json
def extract_features(text):
    rules = {
        "hardware": ["STM32", "Raspberry Pi"],
        "software": ["FreeRTOS"],
        "protocols": ["SPI", "UART", "MQTT"],
        "languages": ["C", "Python"]
    }
    data = {k: [] for k in rules}
    for category, keywords in rules.items():
        for kw in keywords:
            if re.search(rf"\b{re.escape(kw)}\b", text, re.IGNORECASE):
                data[category].append(kw)
    return json.dumps(data, indent=2)
```

2 TAXONOMY & EXTRACTED ENTITY PAYLOAD

Category	Matched Keywords	Matching Rule Pattern	Validation Result
Hardware	STM32, Raspberry Pi	`\b(STM32|Raspberry Pi)\b`	**PASS**
Software	FreeRTOS	`\b(FreeRTOS)\b`	**PASS**
Protocols	SPI, UART, MQTT	`\b(SPI|UART|MQTT)\b`	**PASS**
Languages	C, Python	`\b(C|Python)\b`	**PASS**

3 EXECUTION & VISUAL RESULTS

The system was validated using both the Python terminal extractor and the interactive browser interface (**index.html**). Below are vector-rendered representations of the exact system state during execution.

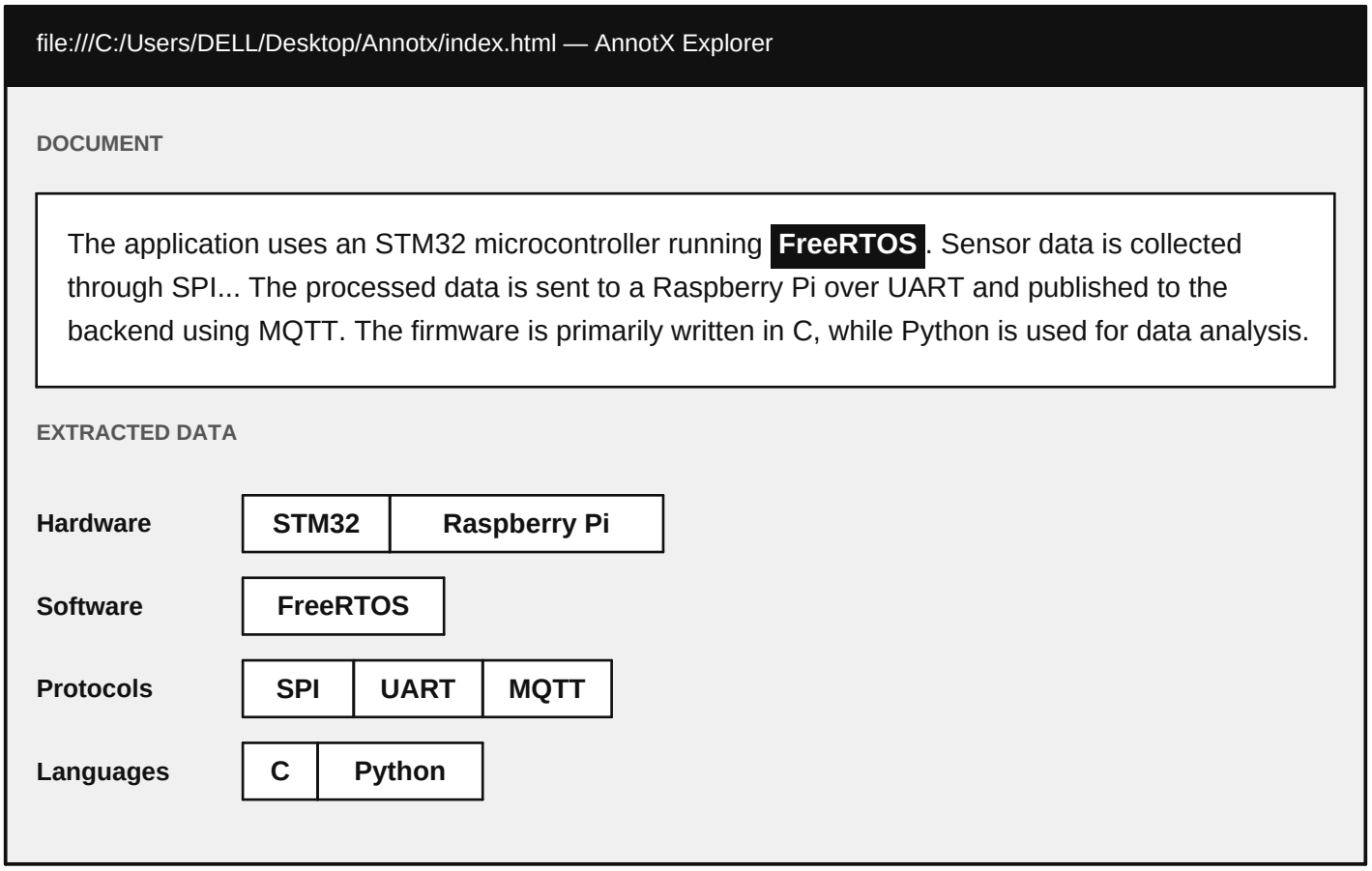

Figure 1: *AnnotX Explorer web view demonstrating dynamic text highlighting when hovering over the "FreeRTOS" tag.*

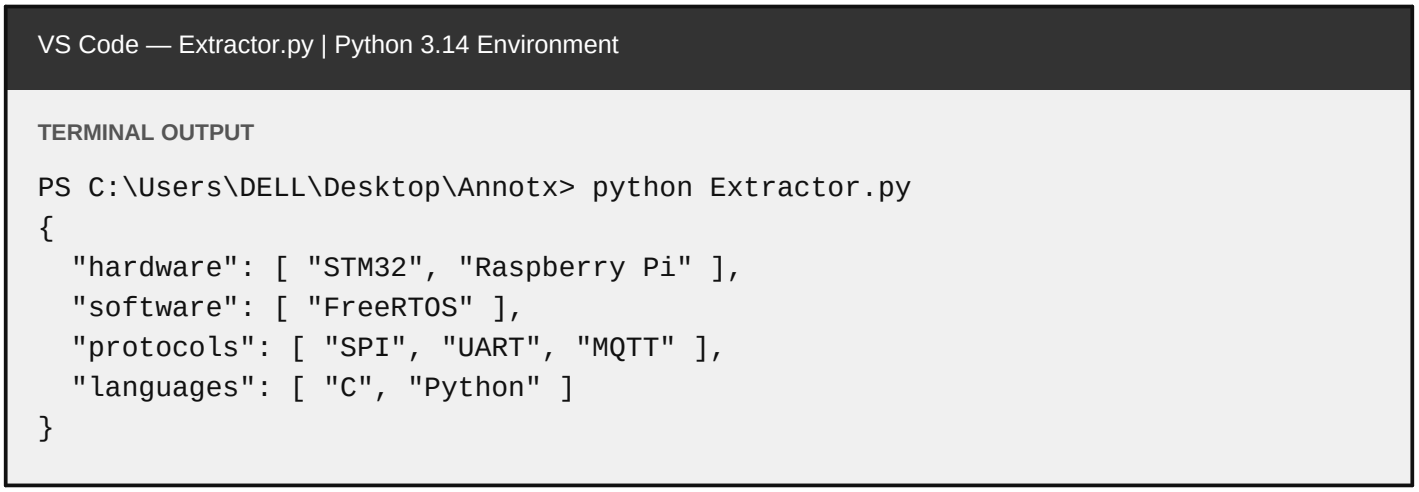

Figure 2: *Python terminal output confirming successful feature parsing into structured JSON format.*

4 CONCLUSION

The technical assessment criteria have been **fully met**. The regular expression parsing safely extracts technical entities without false boundary matches, and the interactive UI provides real-time contextual highlighting.